

Unit 3, Lesson 9: Writing Small Numbers in Scientific Notation



Benchmarks

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.



Learning Targets

- ☐ I can use the laws of exponents to evaluate numerical expressions with a base of 10 and exponents less than 0.
- ☐ I can express small numbers in scientific notation.
- ☐ I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.



3.9.1 Warm-Up: Googol, not Google

A “googol” is a name for a really big number: a 1 followed by 100 zeros.

1. If you square a googol, how many zeros will the answer have? Show your reasoning.



3.9.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the decimal value of each. Then, write the equivalent fractional value.

Power of 10	Decimal Value	Fractional Value
10^0		
10^{-1}		
10^{-2}		
10^{-3}		
10^{-4}		
10^{-5}		

2. What do you notice about the relationship between the power of 10 and its value?

3. Work with your partner to complete the table.

Power of 10	Fractional Value	Place Value of 1
10^0	1	Ones
10^{-1}		Tenths
10^{-2}		
10^{-3}		
10^{-4}		
10^{-5}		



4. Work with your partner to describe how the exponent of the power of ten relates to place value.
5. Based on the pattern in the table, what is the value of 10^{-12} , written as a fraction?



Negative Powers of 10

A base 10 with a negative exponent can be rewritten as a fraction.

$$10^{-a} = \frac{1}{10^a}$$

The value of the fraction relates directly to the place value of the 1 when the quantity is expressed in its equivalent decimal form.



3.9.3 Bring It Together: Applying the Laws of Exponents

The Laws of Exponents can help when rewriting mathematical statements as a power of 10. Sometimes more than one law can be applied.

In the table from the Exploration, $\frac{10^{-11}}{10^{-4}}$ was written as 10^{-7} . Another way to write 10^{-7} is $\frac{1}{10^7}$. Each of these expressions represent the same value.

1. Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$\left(\frac{1}{10^{-4}}\right)^2$		
$(10^{-11})^3 \times 10^{27}$		
$\frac{10^{-13}}{10^{-5}} \cdot 10^8$		

Numbers like one-billionth, also known as a nano, can be written using powers of 10.

2. Write one-billionth, or $\frac{1}{1,000,000,000}$, as a power of 10.
3. One septillionth, 0.000000000000000000000001, is also known as yocto. Write one septillionth as a power of 10.





3.9.4 Guided Instruction: Writing Small Numbers in Scientific Notation

Powers of ten can also be used to express small numbers in scientific notation.

1. A single red blood cell is about 0.000007 meters in diameter.

- a. In what place is the 7 in the value, 0.000007?

- b. Which power of 10 relates to this place value?

- c. Express 0.000007 in scientific notation.

2. Consider the value 0.0000672.

- a. What value between 1 and 10 could be multiplied by a power of 10 to result in 0.0000672?

- b. Which digit is in the ones place of your answer to Part A?

- c. In which place value is that digit in 0.0000672?

- d. Which power of 10 relates to this place value?

- e. Express 0.0000672 in scientific notation.

3. Consider the value 3.4×10^{-9} .

- a. Complete the statements.

10^{-9} relates to the _____ place. When rewriting 3.4×10^{-9} in standard form, the ____ will be in the billionths place, followed by the _____. The places before the 3 will each hold the digit, _____.

- b. Express this value in standard form.



3.9.5 Guided Practice: Writing Small Numbers in Scientific Notation

1. The density of oxygen is 1332 millionths of a gram per cubic centimeter.

- a. Express this value as a decimal in standard form.

- b. Express this value in scientific notation.

2. The Mycoplasmas bacteria is about 2.5×10^{-4} millimeter (mm) in size.

Express this value in standard form.



3.9.6 Collaboration: Comparing Small Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

- a. Complete the table.

Scientific Notation	Standard Form
7.2×10^{-4}	
	0.00000018

- b. Complete the statement to compare the factors being multiplied by powers of 10 in each number that is expressed in scientific notation.

7.2 is _____ times
☐ smaller
☐ larger
 than _____.

- c. Complete the statement to compare the powers of 10 in each number that is expressed in scientific notation.

10^{-4} is _____ times
☐ smaller
☐ larger
 than $10^{\boxed{}}$.

- d. Complete the statement.

The value 7.2×10^{-4} is _____ times
☐ smaller
☐ larger
 than 0.00000018.



3.9.7 Wrap-Up: Ticket Out the Door

1. Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.

988 ten thousandths

$$9.88 \cdot 10^{-4}$$

0.988

$$9.88 \cdot 10^{-5}$$

0.00988

$$9.88 \cdot 10^{-2}$$

0.0000988

$$9.88 \cdot 10^{-3}$$

988 millionths

$$9.88 \cdot 10^{-1}$$

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